

Junfeng Alan Lyu

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Education

Peking University

Sep. 2023 – Jul. 2027

- Major in **Physics** ([School of Physics, PKU](#) 🔗)
Dual degree in **Life Sciences** ([School of Life Sciences, PKU](#) 🔗)
- **CGPA: 3.85/4.00; MGPA: 3.85/4.00; Top 10%**
- **Related Courseworks:**
 - **Physics:** 92 in Topics on nonlinear physics, 93 in Equilibrium statistical physics, 94 in Computational physics, 95 in Electrodynamics, 91 in Modern physics laboratory.
 - **Life Sciences:** 92 in Genetics, 92 in Physiology, 88 in Cell biology, A in Molecular biology, Bioinformatics*, Microbiology*.
 - **Computational Biology:** 95 in Lectures on systems biology, Physical chemistry on cellular and molecular biology*, A in Fundamentals of network science and systems biology, A in Population genetics, 92 in Biology of aging, Foundations and advances in synthetic biology*, Biomathematical modeling*.

* stands for courses in progress.

Standardized Tests

- **101/120** in TOEFL iBT Test (29 in reading, 21 in listening, 24 in speaking, 27 in writing).
My Best Score: 104/120 (30 in reading, 23 in listening, 24 in speaking, 27 in writing)

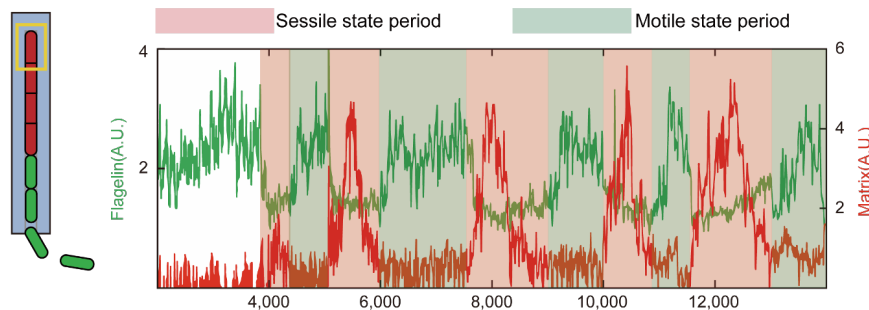
Research

Study on the Colonization Strategies and State Transition Mechanisms of *Bacillus subtilis*

Sep. 2024 – Jan. 2026

Advised by [Prof. Chao Tang](#) 🔗, Center for Quantitative Biology & School of Physics, Peking University

- Improve the microfluidic mother-machine device to enable stable, long-term observation of the motile–sessile state transition of *Bacillus subtilis* mother cells for up to 14 days.
- Clarify the bidirectional (rather than the previously reported monodirectional) switching strategy modulated by environmental conditions, i.e., the bet-hedging strategy.
- Investigate the adaptive advantages of the bet-hedging strategy in fluctuating environments using coarse-grained computational simulations.



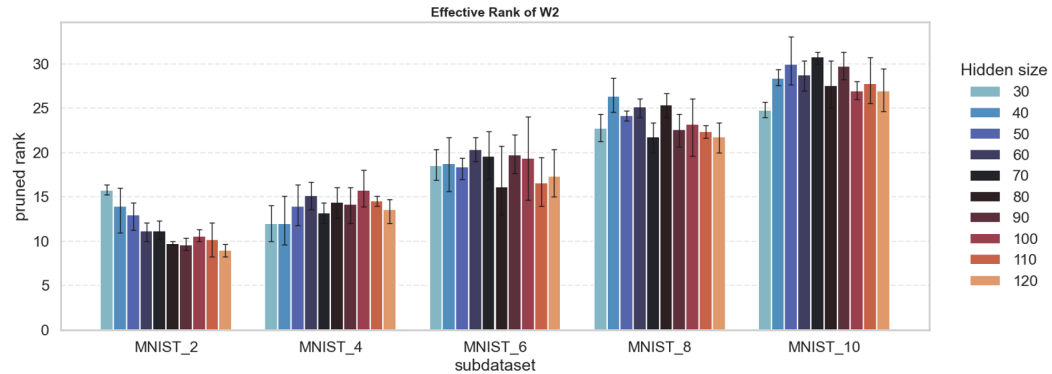
Similarity of the weight space among MLP solution ensemble

Jun. 2025 - Feb. 2026

Advised by [Senior Research Scientist Yuhai Tu](#) 🔗, Center for Computational Biology & Center for Computational Neuroscience, Flatiron Institute and co-advised by [Prof. Qi Ouyang](#) 🔗, Institute for Advanced Study in Physics, Zhejiang University

- Construct the solution ensemble of MLPs with different hidden sizes and label groups of MNIST to study the structure of the weight space shaped by SGD.

- Apply L1 regularization to extract the activated neurons and reveal an universal decay rule of neuron norm and hidden size-independent effective rank of the weight space.
- Apply the orthogonal and permutation alignment of the weight space, evaluating the similarity/performance based on cosine similarity and loss barrier, depicting the basin scenario of the loss landscape in a low-dimensional space based on the alignment task.



Honors and Awards

- **China National Scholarship and Merit Student of Peking University** (2024)
- **China National Scholarship and Merit Student of Peking University** (2023)
- **MCM** — The Mathematical Contest in Modeling
 - Meritorious in MCM2025 (Control Number: #2501436)
 - ? in MCM2026 (Control Number: #2601265)
- **CMC** — The Chinese Mathematics Competitions
 - Provincial second prize in major group-A other than mathematics
- **PKUPT** — Physicists' Tournament among Peking University Students
 - Second prize in PKUPT 2024 (ranked 2/18 teams)
 - The best opponents and commentators awards
- **CPhO** — Chinese Physics Olympiad
 - Provincial first prize in CPhO2022
 - Provincial second prize in CPhO2021
- **IM²C** — The International Mathematical Modeling Challenge
 - Meritorious in IMMC2021 Globe (Control Number: #21451607)
 - Finalist in IMMC2021 China (Control Number: #21042784)
 - Meritorious in IMMC2020 Globe (Control Number: #20063720)
 - Meritorious in IMMC2020 China (Control Number: #20711052)
 - Meritorious in IMMC2019 Globe (Control Number: #19629051)
 - Meritorious in IMMC2019 China (Control Number: #19876444)